OUTPOST SELENE

Junior Astronaut Mission Trainer · Field Guide v3.0

Two destinations. Three training levels. One crew to keep alive.

Operate a lunar or Martian outpost by balancing electrical power, oxygen, water, food production, crew assignments and environmental hazards. Your choices change the outpost and determine whether the crew completes the selected mission.

Start with the tutorial. The six-step interactive tutorial opens automatically when a mission begins. Reopen it any time from Inspect → Interactive tutorial.

Mission setup

Destination	Training level	Duration
Moon · lunar south-polar scenario	Cadet · ages 8–12	7 / 14 / 30 Earth days
Mars · surface outpost scenario	Specialist · ages 13–15	7 / 14 / 30 sols (24.66h each)
	Commander · ages 16–18+	

This guide describes game rules and simplified educational models. It is not an operational spaceflight manual.

1. Your first mission

The solar array begins stowed and OFF. Start the outpost in this order, then check the live objective queue before advancing time.

1. Deploy the solar array

Open Engineering and deploy the array. This is a mechanical action; it does not generate power by itself.

2. Activate solar generation

Turn on generation. The battery stores incoming energy automatically, up to 48 kWh. Solar input depends on destination illumination, panel condition and dust.

3. Start life support

Start the oxygen generator and water recycler. Watch their electrical draw as well as the oxygen and water reserves.

4. Start food production

Assign a crew member to Greenhouse and start the greenhouse. Crops need power and water. Harvest when ready to add food.

5. Check crew assignments

Each role has a limit. Pause a crew member2019s current job before reassigning them when no slot is available.

6. Advance carefully

Use +1H while learning. Larger time jumps pause at hazard-warning milestones so you can respond.

Reading the resource HUD

Power is stored energy (kWh), not a percentage of daily production. Oxygen, water and food are simplified game resource units. The daily balance card estimates energy entering and leaving the system under the current equipment state. If the estimate is negative, plan to pause optional loads or restore generation.

> Do not confuse the battery capacity (48 kWh) with how much energy the solar array can generate in a day.

2. Planet-specific operations

Moon: lunar south-polar scenario

The Moon has no atmosphere to create weather-driven dust storms. In this game, the lunar hazard is an electrostatic dust-adhesion event that can contaminate exposed equipment and reduce solar collection. The simplified light/shadow cycle is a teaching scenario inspired by NASA LRO south-pole illumination mapping; it is not a site-specific LRO time series or live forecast.

Mars: surface outpost scenario

Mars uses a 24.66-hour sol lighting and mission clock; a selected mission length of 7, 14 or 30 means Martian sols, not Earth days. Atmospheric dust fronts can strongly reduce solar input and contaminate panels. A Martian dust front is not the same as a solar-particle radiation event.

Radiation protection

Solar energetic particles are a separate crew-safety hazard. Move the crew into the habitat radiation shelter before the event. The equipment dust-protection shield does not protect crew members from radiation. Surface rover operations are restricted near unsafe hazard windows.

Recovery after a dust event

When conditions are safe, clean the array to reduce dust coverage and repair damaged panels with materials. A mission can be recovered after equipment damage, but a damaged array may produce less power until serviced.

Crew and food management

Greenhouse duty shortens crop growth. Harvest manually when crops mature. Crew Quarters assignment slows fatigue growth. Medical supplies restore health but do not solve a depleted oxygen, water or food supply. Pause and reassign roles when priorities change.

3. Difficulty, controls & science

Training levels

Cadet: generous starting reserves, earlier warnings, milder hazards and guided explanations. **Specialist:** balanced resources, contextual hints and standard warning lead times. **Commander:** tighter reserves, shorter warnings and more severe hazard effects. Difficulty changes the simulation, not just the interface.

Controls

Use the left sidebar to open Engineering, Life Support, Greenhouse, Crew Quarters, Rover Bay, Research Lab, Weather and Objectives. Use the right-hand operations panel for actions. Press 1–6 to select the first six rooms; Space advances one hour. Inspect opens the tutorial, science desk, briefing, field guide, weather, crew, mission log, objectives and settings.

Science sources & model limits

The Science Desk links to NASA LRO south-pole illumination mapping, Curiosity RAD Mars radiation measurements, NASA Veggie plant-growth research and NASA BVAD life-support reference values. The published Curiosity RAD surface reference of about 210 µGy/day is used in Mars missions to calculate a cumulative surface-environment reference counter. It is not an individual crew dose. LRO, Veggie and BVAD references inform the science context and simplified model design. This release does not fetch live NASA APIs or operational forecasts. Event schedules, resource rates, health penalties and crop yields are educational game mechanics, not real mission specifications.

Official reference links

NASA LRO illumination map: https://science.nasa.gov/resource/illumination-map-of-the-moons-south-pole/
NASA Curiosity RAD: https://science.nasa.gov/resource/radiation-measurements-on-mars/
NASA Veggie: https://science.nasa.gov/mission/veggie/
NASA Life Support BVAD: https://ntrs.nasa.gov/citations/20210024855

Accounts and saves

Account creation and saves are local-browser demonstrations. They do not sync across devices and are not backed by a remote authentication server. Guest sessions are not stored.

Mission debrief

At the end of the selected duration, the debrief reports the mission outcome and final crew/resource state. If the crew is lost or a critical supply is depleted at the deadline, the mission fails. Review the mission log to understand which decisions caused the outcome.